SUSTAINABILITY & DISASTER RELIEF

RENOVO

Rubble-to-Brick Press

James Dyson Award 2026 · Project 05

Disaster Relief / Structural Engineering

Regional focus: Turkey · Morocco · Nepal · Central America

Deadline: 15 July 2026

1. PROJECT OVERVIEW

Project Name	RENOVO — Rubble-to-Brick Press
Award	James Dyson Award 2026
Category	Sustainability / Disaster Relief / Structural Engineering
Tag	SUSTAINABILITY
Regional Focus	Turkey, Morocco, Nepal, Central America
Target Cost	Near-zero per brick (energy + binder only)
Deadline	15 July 2026
Status	■ In Progress

2. THE PROBLEM

After major earthquakes — Turkey 2023, Morocco 2023, Nepal 2015 — two crises unfold simultaneously. Communities sit surrounded by millions of tonnes of rubble that must be expensively cleared, while simultaneously importing new building materials at prices they cannot afford. The waste and the need exist side by side, completely disconnected.

Standard rubble management has two options: pay for expensive removal and landfill, or leave rubble in place for years. Some NGOs use manual rubble-sorting for road fill, but no field-deployable system exists to upcycle concrete rubble into structural bricks.

Earthquake	Year	Estimated Rubble Volume	Reconstruction Cost
Turkey (Kahramanmara■)	2023	~210 million tonnes	>$100 billion USD
Morocco (Al Haouz)	2023	~5 million tonnes	>$10 billion USD
Nepal (Gorkha)	2015	~100 million tonnes	>$7 billion USD
Syria (ongoing)	2011–	>900 million tonnes	>$400 billion USD

3. THE SOLUTION

RENOVO is a compact, modular, field-deployable press that crushes concrete rubble, mixes it with geopolymer binder (no kiln required), and compresses it into standard-dimension structural bricks curing at ambient temperature. A full pallet of bricks is produced on-site within 48 hours, using solar power or manual operation — requiring no grid, no fuel, and no imported materials.

- Transportable by small truck — modular, bolts together on-site
- Operable by unskilled labour — minimal training required
- Manufacturable from standard steel sections available in any country
- Dual power: 1.5 kW electric (200 W solar panel) or manual flywheel
- Brick dimensions: ISO 228 × 108 × 76 mm — drop-in compatible with conventional masonry
- Target compressive strength: 15–25 MPa (meets single-storey residential standard)

4. HOW IT WORKS — THREE STAGES

Stage 1 Crushing	Jaw-crusher attachment (electric or manual flywheel) reduces concrete rubble to 5–20 mm aggreg
Stage 2 Mixing	Aggregate enters a mixing chamber with geopolymer binder solution. Sodium silicate activates fly
Stage 3 Compression	Mixture loads into a compression mould and is pressed at 10–15 MPa via hydraulic ram — solar-b

5. GEOPOLYMER BINDER CHEMISTRY

Portland cement was eliminated — it requires controlled curing conditions and does not bind recycled aggregate reliably under field conditions. Alkali-activated geopolymers cure at ambient temperature and achieve equivalent strength within 24 hours.

Component	Role	Source / Availability	Cost

Sodium silicate (Na■SiO■)	Alkaline activator — triggers polymerisation of alumino-silicate precursor	Hardware shops (waterglass), industrial chemical suppliers	~£5–15 / kg
Fly ash (Class F)	Primary geopolymer precursor — provides SiO■ and Al■O■ for network formation	Coal power plant byproduct — free or near-free in most regions	~£0–5 / tonne
GGBS (Ground Granulated Blast-Furnace Slag)	Alternative/supplementary precursor — faster strength gain	Steel industry byproduct — widely available globally	~£30–80 / tonne
Crushed concrete aggregate	Filler / structural matrix — 75% of brick volume	On-site rubble — zero cost	Free
Water	Mix workability adjustment	Local supply	Free

Reaction mechanism: The alkali solution (NaOH + sodium silicate) dissolves Si and Al from the precursor surface, forming Si–O–Al oligomers that condense into a 3D aluminosilicate network — geopolymer gel. This gel fills voids between aggregate particles and hardens progressively at room temperature, reaching structural strength in 24–48 hours and continuing to gain strength over 28 days.

6. MECHANICAL ARCHITECTURE

The press is modular — jaw crusher, mixing drum, and compression press are separate units that bolt together in the field, allowing damaged components to be replaced individually without losing the whole machine.

Module	Function	Power	Key Spec
Jaw Crusher	Reduces rubble to 5–20 mm aggregate	1.5 kW electric or manual flywheel	Throughput: ~50 kg/hr at manual operation
Integral Sieve	Removes rebar and oversized debris	Passive (gravity)	Mesh aperture: 20 mm
Mixing Drum	Combines aggregate + binder at 75:25 ratio	0.5 kW electric or manual paddle	Batch size: ~10 kg per cycle (~5 bricks)
Hydraulic Press	Compresses mixture at 10–15 MPa	Solar battery bank (200 W panel) or manual lever	Mould: ISO 228×108×76 mm
Ejection System	Releases cured brick without damage	Mechanical (spring-loaded)	Cycle time: ~2 min per brick

7. BRICK SPECIFICATION

Dimensions	228 × 108 × 76 mm (ISO standard — drop-in compatible with conventional masonry)
Compressive Strength	15–25 MPa (published geopolymer literature at 75% recycled aggregate, 10–15 MPa press pressure)

Standard Equivalent	Meets requirements for single-storey residential construction in most national codes
Curing Time	24–48 hours ambient — no kiln, no heat, no controlled environment
Aggregate Composition	75% crushed concrete rubble / 25% geopolymer paste by volume
Cost per Brick	Energy + binder only — rubble is free on-site. Binder cost: ~£0.05–0.15 per brick at scale
vs. Commercial Brick	Post-earthquake brick prices spike 300–500% due to demand surge — rubble brick eliminates import cost entirely

8. COMPETITIVE LANDSCAPE

No field-deployable integrated system exists combining rubble crushing, geopolymer mixing, and compression in a single transportable unit. The closest adjacent products:

Category	Existing Products	Gap vs. RENOVO
Compressed Earth Block Presses	CINVA-Ram, Vermeer BP714, Auram 3000	Use virgin soil — not rubble. Not designed for disaster deployment or rubble pre-processing.
Rubble Removal Equipment	Standard excavators, dump trucks	Move rubble to landfill — do not process or reuse it.
NGO Manual Sorting	Manual labour, simple sieves	Produces road fill only — not structural bricks. Extremely low output.
Geopolymer Research	Academic prototypes (Davidovits et al.)	Laboratory scale — not field-deployable, not integrated with crushing stage.

9. MINIMUM VIABLE PROTOTYPE CHECKLIST

Dyson Award requires evidence of iterative development — not a finished machine. The following steps constitute a credible submission prototype:

#	Action	What It Proves	Difficulty
1	Mix geopolymer binder (sodium silicate + fly ash/GGBS) in a paper cup. Cure 24 hr at room temp.	Proves no-kiln binder chemistry hardens without heat — core chemistry claim.	Low — ~£10 materials
2	Hammer-crush a concrete paving slab offcut to 5–20 mm aggregate. Mix with binder at 75:25. Press into any mould (bottle cap, cardboard box). Cure 48 hr.	Physical compressed rubble brick exists — photograph it.	Low–Medium

3	Photograph brick next to a standard brick for scale.	Physical artifact > any render.	Trivial
4	Load-test: stand on it, hit with a hammer. Document qualitatively.	Strength evidence without lab equipment.	Low
5	Build tabletop press: G-clamp + two wooden boards. Photograph in use.	Proves compression achievable with basic tooling.	Low
6	Write 1-page test log: mix ratio, cure time, ambient temp, result.	Design iteration evidence.	Trivial

Minimum time to credible prototype: 3–4 days (1 day sourcing + 1 day mixing + 2 days curing)

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10. JAMES DYSON AWARD — APPLICATION ANSWERS

Project Name (50 chars)

Rubble-to-Brick Press

Quick Description (193 chars)

A compact press that crushes post-earthquake concrete rubble, mixes it with geopolymer binder, and produces standard compressed bricks on-site within 48 hours — no kiln, near-zero cost.

What it does (260 chars)

The press crushes rubble, mixes it with a no-kiln geopolymer binder, and compresses it into standard bricks curing at ambient temperature. A full pallet of structural bricks is produced on-site within 48 hours using solar power or manual operation.

Inspiration (650 chars)

After every major earthquake — Turkey 2023, Morocco 2023, Nepal 2015 — two crises unfold simultaneously. Communities sit surrounded by millions of tonnes of rubble that must be expensively removed, while importing new building materials at prices they cannot afford. The waste and the need exist side by side, completely disconnected. The insight: rubble is not waste — it is unprocessed raw material. Geopolymer chemistry allows concrete aggregate to be rebound without heat, through chemical activation alone. Combining these facts pointed to a field-deployable press that turns disaster debris into reconstruction materials.

What sets it apart (975 chars)

Existing post-earthquake rubble management has two options: pay for expensive removal and landfill, or leave rubble in place for years. Some NGOs use manual rubble-sorting for road fill, but no field-deployable system exists to upcycle concrete rubble into structural bricks. Geopolymer compressed earth block presses exist for new-build construction but use virgin soil, not recycled concrete rubble, and are not designed for disaster-site deployment. RENOVO is the first integrated system combining on-site rubble crushing, geopolymer mixing, and compression in a single deployable unit. It eliminates material transportation costs in both directions: rubble stays on-site and new bricks need not be imported. Post-earthquake brick prices spike 300–500% — rubble bricks cost only the energy and binder to produce.

Future (650 chars)

First prototype to be validated with concrete rubble from demolition sites, testing compressive strength across varying aggregate types and contamination levels. Field trials planned with disaster relief NGOs in earthquake-prone regions. A lightweight flat-pack version will be developed for air-freight delivery in the first 72 hours after a disaster. Long-term, the binder chemistry will be refined to work with broader rubble types including brick, tile, and mixed debris, expanding the addressable waste stream significantly.

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11. FUTURE ROADMAP

Phase	Timeline	Milestone
Phase 0 — Proof of Concept	Now (Jun 2026)	Geopolymer binder sample + hammer-crushed rubble brick. Tabletop G-clamp press. Photograph + test log.
Phase 1 — Prototype	Jul–Sep 2026	Steel-framed tabletop jaw crusher + mixing drum + hydraulic jack press. Test with construction site rubble.
Phase 2 — Field Validation	2027	NGO partnership trial in earthquake-affected region. Systematic compressive strength testing across rubble types.
Phase 3 — Flat-Pack Version	2027–2028	Air-freight-deployable variant. 72-hour post-disaster deployment target.
Phase 4 — Scale	2028+	OEM manufacturing partnership. Binder chemistry extended to brick, tile, mixed debris.